

BUSINESS INTELLIGENCE AND TOOLS

EXPERIMENT - 3

Problem File

Title: Dimensional Modeling and Star Schema Design using Microsoft Power BI

Aim: To design and implement a dimensional data model using Microsoft Power BI by transforming transactional data from the Global Superstore dataset into a Star Schema consisting of Fact and Dimension tables for efficient reporting and analytical processing.

Tools / Software Required

- Microsoft Power BI Desktop
- Global Superstore Dataset.xlsx
- Power Query Editor
- DAX (Data Analysis Expressions)

Dataset Source

Global Superstore Dataset (Kaggle)

<https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset>

Problem Statement

Design and implement a dimensional data model in Microsoft Power BI using the Global Superstore dataset. Identify Fact and Dimension tables, create the required dimension tables and Fact_Sales table, establish a Star Schema with appropriate relationships, and validate the model using visualizations and DAX measures.

Required Tasks

1. Import Orders, Returns, and People worksheets.
2. Identify measurable business events and descriptive attributes.
3. Create Dim_Customer using Customer ID, Customer Name, and Segment.
4. Create Dim_Product using Product ID, Product Name, Category, and Sub-Category.
5. Create Dim_Region using Region, Country, State, and City.
6. Create Dim_Date using Order Date, Year, Quarter, and Month.
7. Remove duplicate records from all dimensions.
8. Create Fact_Sales containing Order ID, Customer ID, Product ID, Region, Order Date, Sales, Quantity, Profit, and Discount.
9. Verify key columns required for relationships.
10. Create Many-to-One relationships between Fact_Sales and all Dimension tables.
11. Arrange the model as a Star Schema.
12. Create Sales by Category, Profit by Region, and Sales Trend by Year visualizations.
13. Create DAX measures for Total Sales, Total Profit, Profit Margin, and Average Order Value.
14. Verify that dimensions correctly filter fact data.

Expected Result: A dimensional model should be created successfully in Power BI with one Fact_Sales table, multiple Dimension tables, appropriate relationships, and DAX measures validated through reports.